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MS4092 COMPREHENSIVE EXAMINATION

Material Engineering

Semester II 2022/2023

Saturday, 4th March 2023, Duration: 30 minutes

Closed Book and Closed Note

Answer all the questions on this problem sheet. State your answer by circling the letter in front of the most correct/accurate answer. Correct answers get additional scores, while wrong answers or questions that are not answered get zero score. If necessary, you may use the back page of the question sheet as 'scratch paper'. Don't forget to fill in your name, student identity number, and/or your signature at the top of EVERY page (in the space provided).

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1. Consider a cylindrical specimen of metal alloy that has a diameter of 8.0 mm. A tensile force of 1000 N produces an elastic reduction in diameter of 2.8 × 10-4 mm. Compute the modulus of elasticity for this alloy, given that Poisson's ratio is 0.30.
   a) 170.5 GPa
   b) 185.5 GPa
   c) 150 GPa
   d) 715 GPa

   transverse strain: $\varepsilon_x = \frac{\Delta d}{d} = \frac{2.8 \times 10^{-4}}{8} = 0,000035$

   longitudinal strain: $\varepsilon_z = \frac{\varepsilon x}{\sigma} = 0,0001167$

   $\sigma = F/A_0 = 19,89\ MPa$ $E = \sigma/\varepsilon_z = 170,5\ GPa$

2. A cylindrical specimen of a titanium alloy having an elastic modulus of 107 GPa and an original diameter of 3.8 mm will experience only elastic deformation when a tensile load of 2000 N is applied. Compute the maximum length of the specimen before deformation of the maximum allowable elongation is 0.42 mm.
   a) 255 mm
   b) 220 mm
   c) 155 mm
   d) 100 mm

   $0,42 \times \frac{107 \times 1000}{\frac{2000}{\frac{\pi}{4} \cdot 3,8 \cdot 3,8}} = 255\ mm$

3. What is the carbon concentration of an iron-carbon alloy for which the fraction of total ferrite is 0.94?
   a) 0.12 wt% C
   b) 0.22 wt% C
   c) 0.42 wt% C
   d) 0.02 wt% C

   $\%Fe = \frac{6,7 - \%Co}{6,7 - 0,022}$

   $\%Co = 6,7 - 0.94(6,7 - 0,022) = 0,42268$

4. In the tensile test, ductility of the materials can be expressed in...
   a) Yield strength
   b) Young Modulus
   c) Percent reduction of area / percent elongation
   d) Ultimate tensile strength

5. What is the crystal structure of iron at room temperature?
   a) Face center cubic
   b) Body center cubic
   c) Hexagonal cubic

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d) Tetragonal cubic

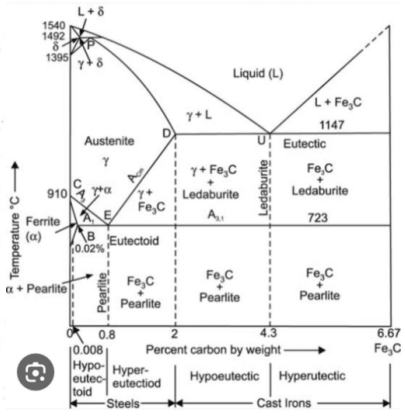
6. What are the chemical bonding exhibits predominantly in polymers?
   a) Van der Walls bonding
   b) Covalent bonding
   c) Ionic bonding
   d) Hydrogen bonding

Primary: Covalent bonds
Secondary: van der waals and hydrogen bonds

7. What kind of test used for measuring hardenability of steel?
   a) Charpy impact test
   b) Brinell test
   c) Rockwell test
   d) Jominy end-quench test

8. The main element contributing to anti-corrosion properties in 316L stainless steel is...
   a) Carbon
   b) Iron
   c) Nickel
   d) Chromium

9. AISI 1020 materials is heated to 900°C, then directly immersed in water. What is the phase of the materials in room temperature?
   a) Pearlite
   b) Ferrite
   c) Austenite
   d) Martensite



10. Which materials has no ductile to brittle transition temperature?
    a) Low Carbon Steel
    b) Aluminum alloy → struktur FCC
    c) Medium steel
    d) Chromium Alloy

11. The following properties of thermoplastic polymers is superior to the properties of steel
    a) Higher tensile strength
    b) Higher stiffness
    c) Higher ductility
    d) Higher yield strength

12. What is the nominal carbon content in of AISI 1010 at room temperature?
    a) 10%
    b) 1%
    c) 0.1%
    d) 0.01%

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